Ramon McDargh-Mitchell

Full-Stack Software Engineer & ML Developer
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Education

University of California San Diego (UCSD) Expected June 2027
Double Major: B.S. Computer Science & B.S. Cognitive Science (Machine Learning & AI Specialization) GPA: 3.6
Relevant Coursework: Data Structures, Algorithms, Machine Learning, Neural Networks, Statistical Analysis

Technical Skills

Core Technologies: Next.js 16, React 19, TypeScript, Python, C++, Node.js, SQL
Backend & Databases: PostgreSQL, Supabase, MySQL, Prisma ORM, REST APIs, GraphQL, NextAuth, Better Auth
Cloud & DevOps: Docker, Linux, GitLab CI/CD, Vitest, GCP Cloud Run, AWS S3, Vercel, Oracle Cloud, Upstash Redis, Sentry
Drones & Hardware: VOXL, voxl-portal, ADB Shell, Camera Calibration, Flight Log Analysis, Board Flashing & Bench Testing
ML & AI: XGBoost, Random Forest, YOLOv8, OpenCV, Bayesian Forecasting, Scikit-learn, Pandas, NumPy, Matplotlib
Game Dev & VR: Unreal Engine 5, Virtual Reality Development, HCI, Simulation
Tools & Integrations: Git, Stripe, Resend, Shopify API, Zod, TanStack, Webhook Integration, GSAP

Professional Experience

ModalAI | *Software Engineer* June 2026 - Present
- Ship features and bug fixes (650+ commits) for the internal platform that runs drone production, purchasing, and inventory
- Linked work orders to Shopify and stock orders; built a multi-build planner, alternate parts, and low-stock ranking algorithms
- Built demand forecasting and measured vendor lead times from order history, turning them into reorder dates and buy plans
- Built test analytics for board flashing and flight logs (lot and serial tracking, yield, pass rates) and moved it to PostgreSQL
- Calibrated drone cameras and validated flights on VOXL hardware using voxl-portal, ADB shell, and Linux tooling

Qualcomm Institute at UCSD | *VR Research Intern* Fall 2025 - Present
- Developing AI-powered virtual reality avatars using Unreal Engine 5 for educational classrooms and museums
- Contributing AI object detection to VR SmartClassroom and NanoVerseVR for guided tutorials and nanoscale exploration
- Collaborating with faculty leaders on immersive technology research blending game development, ML, and HCI

rvector.dev | *Founder & Full-Stack Developer* Spring 2025 - Present
- Founded web development company serving real clients with full-stack solutions using Next.js 16, React 19, TypeScript
- Built comprehensive business platforms with admin dashboards, project management, payment processing, real-time data sync
- Deployed production applications with cloud infrastructure, monitoring, and CI/CD pipelines gaining me 3+ clients

Key Projects

Ascend - Hackathon & Ideathon Platform Winter 2025 | *Visit Site*
- Built full-featured competition platform for organizations and companies using Next.js 16, React 19, Prisma ORM, AWS S3
- Implemented end-to-end workflow: organization management, event creation, team formation, judging, and live leaderboards
- Helping 3+ organizations save 10+ hours each by eliminating need to build custom infrastructure for competitions

ML Research: Empirical Comparison of AI Models Fall 2024 | *View Project*
- Conducted comprehensive empirical study comparing XGBoost, Random Forest, and Neural Networks following NeurIPS format
- Built ML pipeline evaluating 3 classification and 2 regression datasets across multiple train-test splits with 10-fold CV
- Published research paper showing XGBoost outperformed on 2/3 classification and 1/2 regression datasets

Custom YOLOv8 Object Detection Model Spring 2025 | *View Project*
- Trained domain-specific YOLOv8 model achieving 79.4% fitness score on 351 hand-labeled images
- Built complete ML pipeline: video recording, frame extraction, annotation, training, and deployment to detect personal objects

Technical Achievements & Leadership

- **Demand Forecasting Research:** Analyzed 1,290 orders and 23k stock movements to recommend a Bayesian model for ModalAI
- **Payment Integration:** Implemented complex payment flows including subscriptions, webhooks, and order management
- **UCSD Table Tennis:** Key player securing 1st place for JV team, defeating UCLA
- **HSF Scholar:** Was awarded the prestigious HSF Scholar award for my academic excellence
- **Upperclassmen Excellence in Mathematics:** Chosen by El Dorado High School math faculty for excellence in math

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Portfolio: rvector.dev | **GitHub:** github.com/RamonsArchive